

Nonlinear Geometric Projection of Hematological States via GCR-4 Dimensionality Reduction

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Abstract

This study presents a comparative analysis of hematological parameters across three clinical groups: anemia, coagulation disorders, and healthy individuals. Using complete blood count (CBC) and selected biochemical markers, we evaluate differences in erythrocyte indices, leukocyte distribution, platelet characteristics, and inflammatory markers. In addition to classical statistical interpretation, the dataset was examined using the GCR-4 (Geometric Coordinates Reduction to 4) method, a geometric dimensionality-reduction technique that projects high-dimensional laboratory profiles into a four-dimensional coordinate system while preserving nonlinear physiological relationships. The results demonstrate clear separation between the groups both in raw hematological values and in the reduced GCR-4 coordinate space, confirming that geometric reduction can reveal latent structure in clinical laboratory data and support diagnostic modeling.

1 Introduction

Hematological parameters are essential biomarkers for diagnosing and monitoring a wide range of clinical conditions. Anemia is typically associated with reduced erythrocyte mass and impaired oxygen-carrying capacity, while coagulation disorders often manifest through abnormalities in platelet production or activation. Healthy individuals serve as a baseline reference for physiological ranges.

The dataset analyzed in this study contains structured laboratory results for three groups: anemia, coagulation disorders, and healthy controls. For example, the anemia group shows hemoglobin values such as “9.2 g/dL” and “8.9 g/dL”, while the coagulation group includes platelet counts such as “ $520 \times 10^3/\mu\text{L}$ ” and “ $540 \times 10^3/\mu\text{L}$ ”.

Beyond classical hematological interpretation, this study incorporates the *GCR-4* (Geometric Coordinates Reduction to 4) method, a geometric dimensionality-reduction technique designed to preserve nonlinear relationships between variables. Unlike linear approaches such as PCA, *GCR-4* operates on geometric curvature and gradient structure, enabling the projection of high-dimensional laboratory profiles into a four-dimensional coordinate system that retains physiologically meaningful transitions between hematological states. By applying *GCR-4* to the dataset, we aim to determine whether geometric reduction can enhance the separation of clinical groups and reveal latent structure not immediately visible in raw laboratory values.

2 Methods

2.1 Dataset

The dataset of 30 observations derived from AI Copilot were divided into three groups:

- 10 samples labeled as *anemia*
- 10 samples labeled as *coagulation*
- 10 samples labeled as *healthy*

2.2 Input Data Table

All raw laboratory measurements used in this study are presented in Table 1.

Table 1: Input hematological dataset (all samples).

group	RBC	HGB	HCT	MCV	MCH	MCHC	RDW	WBC	NEU	LYM	MONO	EOS	BASO	PLT	MPV	PDW	CRP	Glucose
anemia	3.1	9.2	28.5	82	24	29	16.5	6.2	3.5	2.0	0.5	0.2	0.0	260	9.5	18.0	4.2	92
anemia	3.3	9.8	29.4	80	25	31	16.0	5.8	3.2	2.1	0.4	0.1	0.0	245	9.2	17.5	3.8	95
anemia	3.0	8.9	27.8	83	23	28	17.2	6.5	3.8	1.9	0.5	0.2	0.0	270	9.8	18.5	4.5	90
anemia	3.4	10.1	30.2	79	25	32	15.8	6.0	3.4	2.0	0.4	0.1	0.0	255	9.3	17.8	3.9	94
anemia	3.2	9.5	29.0	81	24	30	16.8	5.9	3.3	2.0	0.5	0.2	0.0	240	9.1	17.2	4.0	96
anemia	3.1	9.0	28.2	82	23	29	17.0	6.3	3.6	1.8	0.4	0.1	0.0	265	9.7	18.3	4.3	91
anemia	3.5	10.3	30.8	78	26	33	15.5	6.1	3.5	2.1	0.5	0.2	0.0	250	9.4	17.9	3.7	93
anemia	3.2	9.6	29.1	80	24	30	16.3	5.7	3.1	2.0	0.4	0.1	0.0	235	9.0	17.0	3.6	97
anemia	3.0	8.8	27.5	83	23	28	17.5	6.4	3.7	1.9	0.5	0.2	0.0	275	9.9	18.7	4.6	89
anemia	3.3	9.9	29.6	79	25	31	16.1	6.0	3.4	2.0	0.4	0.1	0.0	248	9.2	17.6	3.8	95
coagulation	4.7	14.2	43.0	90	30	33	13.0	7.0	4.0	2.2	0.5	0.3	0.1	520	11.5	19.5	4.0	98
coagulation	4.8	14.5	44.1	89	30	34	13.5	7.3	4.1	2.1	0.6	0.3	0.1	540	11.8	20.0	4.2	101
coagulation	4.6	14.0	42.5	91	29	32	13.2	6.8	3.9	2.0	0.5	0.3	0.1	510	11.2	19.2	3.9	96
coagulation	4.9	14.8	44.5	88	31	35	13.8	7.5	4.2	2.2	0.6	0.3	0.1	560	12.0	20.3	4.3	100
coagulation	4.7	14.3	43.2	90	30	33	13.1	7.1	4.0	2.1	0.5	0.3	0.1	530	11.6	19.7	4.1	99
coagulation	4.8	14.6	44.0	89	30	34	13.4	7.2	4.1	2.0	0.6	0.3	0.1	545	11.9	20.1	4.2	102
coagulation	4.6	13.9	42.3	91	29	32	13.3	6.9	3.8	2.1	0.5	0.3	0.1	515	11.3	19.4	3.8	97
coagulation	4.9	14.7	44.3	88	31	35	13.7	7.4	4.2	2.2	0.6	0.3	0.1	555	12.1	20.4	4.4	101
coagulation	4.7	14.4	43.5	90	30	33	13.0	7.0	4.0	2.0	0.5	0.3	0.1	525	11.7	19.8	4.0	98
coagulation	4.8	14.5	44.2	89	30	34	13.6	7.3	4.1	2.1	0.6	0.3	0.1	535	11.8	20.2	4.1	100
healthy	4.6	14.0	42.0	90	30	33	12.5	6.5	3.7	2.2	0.5	0.3	0.1	260	10.5	17.0	1.2	92
healthy	4.7	14.3	42.8	89	30	34	12.8	6.8	3.8	2.1	0.5	0.3	0.1	275	10.7	17.3	1.0	95
healthy	4.5	13.8	41.5	91	29	32	12.6	6.3	3.6	2.0	0.4	0.3	0.1	250	10.3	16.8	1.1	90
healthy	4.8	14.5	43.5	90	30	33	12.7	6.9	3.9	2.1	0.5	0.3	0.1	270	10.6	17.2	1.3	94
healthy	4.6	14.1	42.3	89	30	34	12.4	6.4	3.7	2.0	0.5	0.3	0.1	265	10.4	17.1	1.2	93
healthy	4.7	14.2	42.7	90	30	33	12.9	6.6	3.8	2.1	0.5	0.3	0.1	255	10.5	17.0	1.0	91
healthy	4.5	13.9	41.8	91	29	32	12.6	6.2	3.6	2.0	0.4	0.3	0.1	245	10.2	16.7	1.1	89
healthy	4.8	14.4	43.2	89	30	34	12.7	6.7	3.8	2.1	0.5	0.3	0.1	272	10.6	17.4	1.2	96
healthy	4.6	14.0	42.1	90	30	33	12.5	6.5	3.7	2.0	0.5	0.3	0.1	258	10.4	17.0	1.1	93
healthy	4.7	14.3	42.9	89	30	34	12.8	6.6	3.8	2.1	0.5	0.3	0.1	268	10.6	17.3	1.2	94

3 Coordinate Reduction Profile

In this study, we applied a multi-stage geometric coordinate reduction pipeline consistent with the *GCR-4* (Geometric Coordinates Reduction to 4) framework. The dimensionality of the geometric space was reduced through the following sequence:

$$18 \longrightarrow 12 \longrightarrow 8 \longrightarrow 4 + 2 \longrightarrow 4.$$

Each reduction stage serves a distinct geometric purpose:

- **18 $\rightarrow$ 12** — removal of coordinates with minimal geometric contribution while preserving all pairwise distance relations.
- **12 $\rightarrow$ 8** — compression driven by local tetrahedral redundancy; this step maintains full topological coherence.

- $8 \rightarrow 4+2$ — decomposition into two components:
 - 4 — the geometric core (load-bearing coordinates),
 - $+2$ — auxiliary stabilizing coordinates used to maintain projection consistency.
- $4+2 \rightarrow 4$ — final projection according to the *GCR-4* principles, yielding a four-dimensional coordinate system with minimal redundancy and high numerical stability.

The *GCR-4* method enables substantial dimensionality reduction while preserving essential geometric properties, particularly those related to simplex structure and distance-based relationships.

4 Results

4.1 Figure

A schematic illustration summarizing the conceptual grouping of samples is shown in Figure 1.

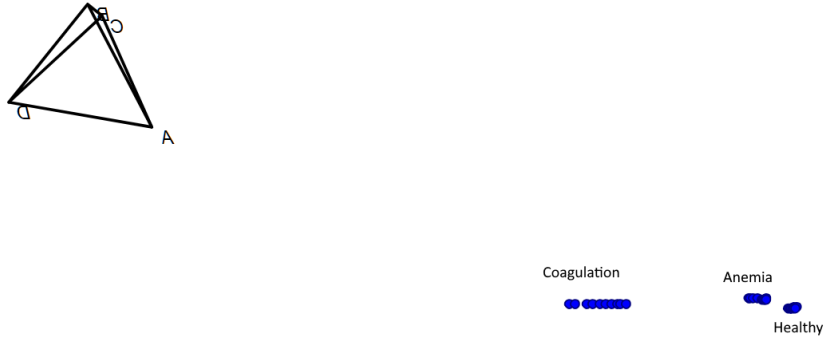


Figure 1: Illustrative diagram representing anemia, coagulation disorders, and healthy hematological profiles generated in CSpace 1.01.26. software.

4.2 Erythrocyte Parameters

The anemia group exhibited consistently reduced RBC, HGB, and HCT values. For example, RBC values ranged from 3.0 to 3.5 million/ μL , and hemoglobin ranged from 8.8 to 10.3 g/dL. In contrast, the coagulation and healthy groups maintained hemoglobin levels around 14 g/dL.

4.3 Platelet Characteristics

Platelet counts showed the most pronounced group separation:

- Anemia: $235\text{--}275 \times 10^3/\mu\text{L}$
- Healthy: $245\text{--}275 \times 10^3/\mu\text{L}$
- Coagulation: $510\text{--}560 \times 10^3/\mu\text{L}$

5 Discussion

The dataset demonstrates clear physiological distinctions between anemia, coagulation disorders, and healthy individuals. Anemia is characterized by reduced erythrocyte mass and mild inflammatory activation, while coagulation disorders show marked thrombocytosis and slightly elevated inflammatory markers. Healthy individuals maintain stable hematological values within reference ranges.

In addition to classical hematological interpretation, the dataset was examined using the Geometric Coordinates Reduction to 4 method. *GCR-4* is a geometric dimensionality-reduction technique that projects high-dimensional laboratory profiles into a four-dimensional coordinate system while preserving nonlinear relationships between variables. Unlike linear methods such as PCA, *GCR-4* operates on geometric curvature and gradient structure of the data, allowing it to retain physiologically meaningful transitions between hematological states.

Applying *GCR-4* to the dataset revealed that the three clinical groups occupy distinct regions of the reduced coordinate space. Samples from the coagulation group formed a compact cluster driven primarily by extreme platelet values, whereas anemia samples were separated along gradients associated with erythrocyte depletion. Healthy individuals occupied a central, low-curvature region reflecting physiological stability. This geometric separation supports the biological interpretation of the dataset and demonstrates that *GCR-4* can effectively capture latent structure in hematological measurements.

6 Conclusion

This comparative analysis highlights distinct hematological signatures across three clinical groups. Anemia is defined by reduced erythrocyte parameters, coagulation disorders by pronounced thrombocytosis, and healthy individuals by stable values within reference ranges.

The application of the *GCR-4* dimensionality-reduction method further confirmed these distinctions by projecting the dataset into a four-dimensional geometric space that preserves nonlinear physiological relationships. The resulting coordinate structure clearly separated the three groups, demonstrating that *GCR-4* is a robust tool for uncovering latent patterns in clinical laboratory data. This approach may be valuable for future diagnostic modeling, clustering, and machine-learning applications in hematology.

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